An Anomaly-Free Representation Learning Approach for Efficient Railway Foreign Object Detection

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***Abstract*—Applying machine vision to facilitate railway anomaly detections faces a grand challenge in that anomalous samples for model training are insufficient due to their infrequent occurrence and wide diversity. An anomaly-free representation learning approach (ARLA) is developed in this article to realize a machine vision-powered railway foreign object detection (RFOD) that does not rely on anomalous samples. The ARLA consists of two components, a memory-suppress diffusion network module and a contrastive dissimilarity network. The former network module well considers the diversity of normal patterns and reconstructs high-fidelity normal images. The latter network module enables image-level and pixelwise foreign object detections based on well-defined dissimilarity scores and distance maps. The ARLA realizes an efficient RFOD, which leverages only normal images in training and does not compromise the detection performance at the inference stage. Improvements offered by ARLA in terms of pixelwise detection performance and model complexity against two groups of benchmarks have been consistently observed based on computational studies using the railway dataset.**

***Index Terms*—Anomaly-free, contrastive learning, diffusion models, image analytics, railway scenes.**

I. Introduction

IN RAILWAY contexts, ensuring safety and operational efficiency heavily relies on the detection of anomalies, particularly foreign objects on rail tracks. Manual inspection often falls short in meeting the demands of high precision and efficiency necessitated by advancements in industrial intelligence. Recent attention has been directed toward anomaly detection in computer vision using machine/deep learning models. Various methodologies aim to classify normal and anomalous images within railway settings. Despite these efforts, challenges persist in achieving reliable and effective foreign object detection due to the specific characteristics of railway anomalies. The following discussion delves into these challenges in detail.

1) *Limited accessibility of field data:* The accumulation and retention of extensive data hold the potential for enhancing analysis and facilitating immediate insights using progressive data-driven methods. However, the lack of publicly accessible field data presents a significant barrier. Concerns on data privacy, intellectual property rights, and logistical challenges of managing and sharing valuable resources often lead to the nondisclosure of datasets. Consequently, gaining access to authentic anomalous data becomes a serious impediment, hindering the development of effective anomaly detection techniques.
2) *Limited availability of anomalous images:* Anomalies in industrial environments are often sporadic and infrequent, resulting in a scarcity of anomalous images for training and evaluation purposes. Although a subset of defective images can be obtained, the inherent imbalance between anomalous and normal instances poses a significant challenge. Such scarcity underscores the importance of innovative approaches that can effectively utilize limited data resources and address class imbalance issues.
3) *Imprecise outcomes of anomaly detection:* Precise anomaly detection requires distinguishing between normal and anomalous instances at both image and pixel levels. However, obtaining pixelwise annotations for anomaly detection presents formidable challenges. The manual annotation process is labor-intensive, time-consuming, and prone to human errors, introducing biases and inaccuracies in the labeling process. Such imprecision undermines the reliability and effectiveness of anomaly detection systems in real-world applications, where accurate anomaly localization is critical for timely intervention and decision-making.

This study focuses on the field of railway foreign object detection (RFOD) and introduces a novel approach called

Manuscript received 29 June 2023; revised 27 September 2023, 10 February 2024, and 17 May 2024; accepted 4 June 2024. Date of publication 4 July 2024; date of current version 7 October 2024. This work was supported in part by Shenzhen-Hong Kong-Macau Science & Technology Category C Project (SGDX20220530111205037), in part by the National Natural Science Foundation of China (72371215 and 72032005), in part by the Research Grant Council of Hong Kong (11200621 and 11201023), and in part by the Hong Kong Innovation and Technology Commission (InnoHK Project CIMDA) and Hong Kong Institute of Data Science under Project 9360163. Paper no. TII-23-2378. *(Corresponding authors: Zijun Zhang; Min Xie)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TII.2024.3413965.

Digital Object Identifier 10.1109/TII.2024.3413965

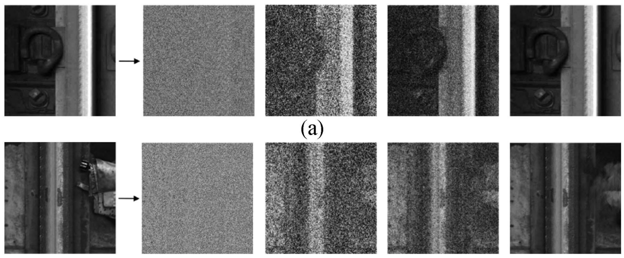


Fig. 1. Motivation of ARLA. (a) Normal rail image is well reconstructed by memory-suppress diffusion. (b) Anomalous rail image is reconstructed with the normal patterns recorded in memory-suppress diffusion.

the anomaly-free representation learning approach (ARLA). It addresses the aforementioned challenges by incorporating anomaly rejection mechanisms into the learning process. One of the key advantages of ARLA is that its training is free of anomalous samples at the representation learning stage. The anomaly rejection mechanism in ARLA ensures that the learned representations predominantly capture normal or nonanomalous patterns. This approach effectively minimizes the influence of outliers or noises in the training data, resulting in more robust, and reliable representations. Moreover, during the testing phase, ARLA offers versatility by providing both image-level and pixelwise detection results. Such a capability enables ARLA to offer comprehensive insights into the presence and location of anomalies contained in railway images. Image-level detection results provide an overall assessment of anomaly presence, whereas pixelwise detection results offer finer-grained information, pinpointing the exact locations of anomalies within the image. This comprehensive detection capability enhances the utility of ARLA in real-world applications, where precise anomaly localization is crucial for effective decision-making and intervention. Fig. 1 provides a visual depiction of our motivation in the context of anomalous railway scenes.

Our main contributions are summarized as follows.

1) We present a pioneering attempt for formulating RFOD as an anomaly-free representation learning process, which aims to address the challenge of limited anomalous data in railway contexts.
2) A novel memory-suppress diffusion process that captures the prototypical patterns of normal data in ARLA is introduced. By quantizing code memories, we enhance the reconstruction of images and prevent excessive model generalization, thereby improving the accuracy of RFOD.
3) A novel dissimilarity loss function is proposed to develop the contrastive dissimilarity network in ARLA. It ensures the discriminative power between the input and its reconstruction, enabling both image-level and pixelwise detection results.
4) The ARLA offers a new state-of-the-art performance for RFOD, which is verified via computational experiments conducted on a collected real dataset. Ablation studies are also conducted to justify the effectiveness and superiority of each component in ARLA.

II. Related Work

In the field of RFOD, extensive research has been conducted on the task of classifying and segmenting anomalous instances within images. This section provides a comprehensive review of the approaches developed specifically for image-level anomaly detection in railway infrastructure. Additionally, we explore the methods that have been generalized to handle pixelwise anomaly detection on widely recognized public datasets. By examining both specialized techniques for railway infrastructure and more general methods, we gain valuable insights into ensuring the safety and reliability of intricate railway systems.

A. Anomaly Detection in Railway Infrastructure

Due to the complexity and numerous subcomponents within railway infrastructure, there exists a broad spectrum of research areas dedicated to anomaly detection in this domain. Existing methods primarily focus on identifying anomalous scenes related to rail tracks, rolling stock, and catenary systems [1]. Maintenance tasks for railway infrastructure can be broadly classified into two main categories, surface defect detection (such as cracks and corrugations on rails [2], [3] and defects on fasteners [4]) and anomaly inspection (such as broken/missing parts [5], [6], foreign objects/obstacles [7], and anomalies in components [8], [9]). The image data employed in these research endeavors are often acquired through custom systems and meticulously hand-labeled, thus limiting their public availability.

Considering the scale and availability of the image data, some studies leverage the third-party data to pretrain the model and adopt transfer learning approaches [2], [10]. Pretrained models serve as the initial learning point, preserving computational resources and enhancing overall detection performance.

During the inference stage, many studies focus on image-level anomaly detection in railway infrastructure while neglecting the localization of anomalies. To address this limitation, segmentation tasks aim to precisely assign pixels within input images to anomalous regions [4], [11]. The output is a segmentation map with the same size as the input image, which facilitates subsequent diagnosis and maintenance. Nevertheless, the segmentation of anomalies is usually implemented in a supervised manner, calling for more advanced data-driven solutions to perform pixelwise anomaly detection when limited anomalous image data is available.

B. Anomaly Detection via Deep Generative Models

As a typical type of method for anomaly detection, deep generative models strive to compress and reconstruct the normal images and detect potential anomalies by evaluating the pixel differences between the inputs and their reconstructions. Autoencoder-based (AE-based) [12] methods and generative adversarial network-based (GAN-based) methods [13], [14] have been widely employed in industrial image analysis, exhibiting promising results in anomaly localization. Typically, the learning process of these methods follows a semisupervised manner, assuming that only normal image representations are learned. During testing, the models struggle to reconstruct anomalous images as accurately as normal ones, providing an opportunity for anomaly detection. In addition to commonly used distance metrics [13], [15], reconstruction probability [12] and likelihood score [16] are defined as supplementary anomaly measures.

AE-based methods possess a strong theoretical foundation, but designing an effective loss function remains challenging, resulting in suboptimal reconstructions. GAN-based methods perform well in various applications, yet they face issues such as training difficulties, vanishing gradients, model collapse, and limited diversity in generated outputs. In summary, generating high-quality images for comparisons is a daunting task, as reconstructing sharp edges and complex texture characteristics often proves challenging, leading to a high number of false abnormal alarms [17].

Denoising diffusion probabilistic models (DDPMs) [18], originating from probabilistic likelihood estimation methods, have recently emerged as powerful tools in computer vision tasks, particularly in generative modeling. DDPMs excel in generating high-quality and diverse samples through forward and reverse diffusion stages [19]. They incorporate a vast amount of images and utilize multiscale representations from the diffusion decoder [20]. Compared with AE- and GAN-based methods, DDPMs offer both tractability and flexibility in analytically evaluating and fitting arbitrary data structures. They possess advantages over alternative generative models when dealing with smaller datasets, as they provide improved sample quality and a stable training scheme. Furthermore, the continuous advancements in the sampling speed of DDPMs (requiring significantly fewer sampling steps, up to hundreds of times faster) demonstrate their potential in terms of time efficiency and computational cost [21], [22].

C. Anomaly Detection via Contrastive Representation Learning

The ability to learn latent representations of raw input data establishes a connection to the broader domain of representation learning. As one of the powerful approaches in self-supervised learning, contrastive representation learning (CRL) has shown promising results in capturing shared features among similar images while distinguishing differences among dissimilar images [23].

The main idea of CRL is to define a similarity distribution that maps the similar pairs close and the dissimilar samples far apart in the embedding space. Unlike the supervised approaches, CRL defines similarity based on the data itself, thereby alleviating the reliance on a labeled dataset with a substantial number of anomalies. Nevertheless, if additional labels are available, CRL can be flexibly integrated into a supervised detection framework [24]. As a result, CRL offers a straightforward yet potent means to learn representations in both supervised and self-supervised settings.

Previous research has developed notable CRL-based methods, including contrastive predictive coding (CPC) [25], augmented multiscale deep infoMax (AMDIM) [26], and momentum contrast (MoCo) [27]. These methods aim to learn features for effectively distinguishing training samples from other samples rather than focusing on capturing every detail of training samples, which is a major difference from the GAN-based methods. Progressive methods enhance their robustness by relying less on negative pairs through asymmetric structures [28], [29] or novel objective functions [30].

So far, CRL-based methods continue to be an active area of research, and we anticipate that these methods will pave the way for future advancements in pixelwise anomaly detection.

TABLE I
TABLE OF NOTATIONS

Notation	Description
x	Input image
X	Image features
q, p	Density distribution
β	Variance scheduler
t, T	Time step
M	code memory
w, v	Corresponding weights
h	Projector output
C	Correlation map
$\mathcal{E}(x)$	Dissimilarity score
M_s, M_d	Distance map
$\mathcal{L}_{vlb}$	Variational lower bound loss
$\mathcal{L}_s$	Noise loss
$\mathcal{L}_d$	Contrastive dissimilarity loss

III. ARLA Framework

This section presents the ARLA, a novel framework designed to generate detailed anomaly maps in anomalous railway scenes. The overview of ARLA is illustrated in Fig. 2. Initially, we explain the memory-suppress diffusion module and its role in reconstructing input images. Subsequently, we introduce the contrastive dissimilarity network, which measures anomaly maps between input and reconstruction, providing image-level and pixelwise detection results. Finally, we define the training mechanism and illustrate the testing procedure, which handles different anomalies in railway scenes.

Table I lists the main notations involved in Section III.

A. Memory-Suppress Diffusion Module

The proposed memory-suppress diffusion module comprises three essential steps. First, the noise encoding step follows the basic diffusion process in DDPM to generate a sequence of noise-perturbed images [18]. Next, a set of code memories obtained from the previous step is integrated to establish consistent representations of normality in the normality memorizing step. Finally, the denoise memory-suppress sampling step continuously reconstructs the normal railway images from the Gaussian noise input using memory-suppression techniques.

1) Noise Encoding: The goal of the noise encoding process in DDPM is to produce a sequence of noisy images $\{x_1, x_2, \ldots, x_T\}$ given an input image x_0 from the real image dataset with density distribution $x_0 \sim q(x_0)$. The noise level of the image is steadily increased in T steps, which follows the Markovian process

$$q\ (x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t\mathbf{I}\right)\ \forall t \in \{1, \ldots, T\} \tag{1}$$

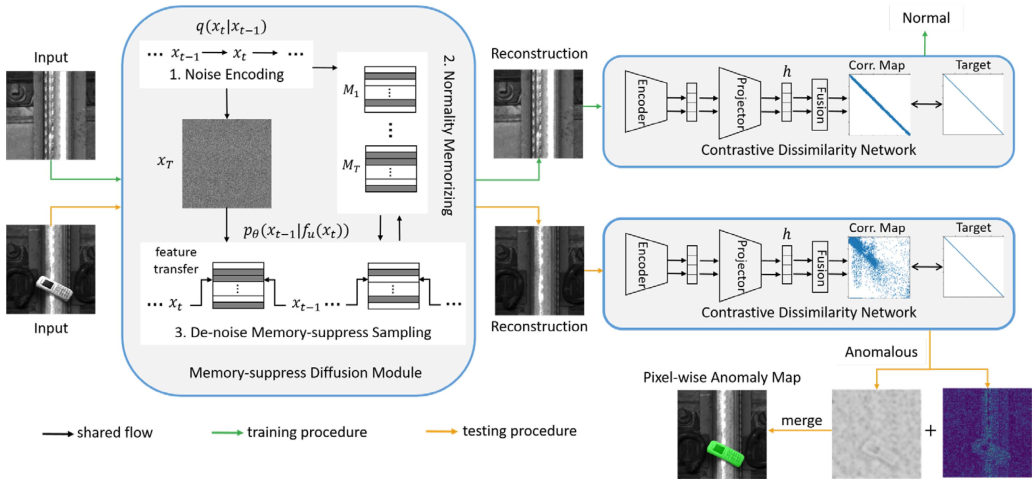


Fig. 2. Overall architecture of the proposed ARLA.

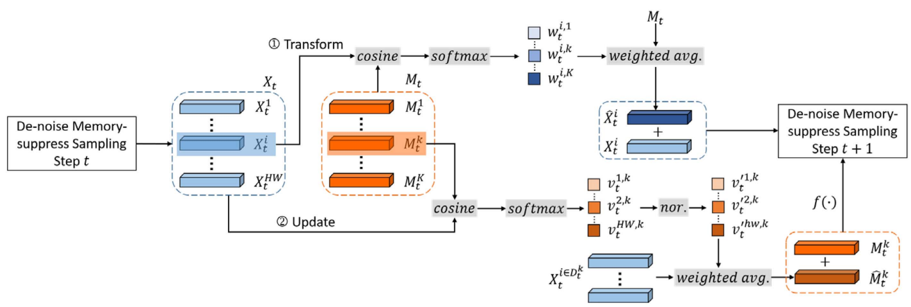


Fig. 3. Architecture of the memory-suppress diffusion module.

$$q\left(x_{1:T}|x_0\right) = \prod_{t=1}^{T} q\left(x_t|x_{t-1}\right) \tag{2}$$

where the step sizes are dominated by a variance scheduler $\{\beta_t \in (0,1)\}_{t=1}^{T}$. Usually, a larger update step is obtained when the image becomes noisier and thereby $\beta_1 < \beta_2 < \cdots < \beta_T$. $\mathbf{I}$ is the identity matrix with the same dimension as x_0. The normal distribution $\mathcal{N}$ of mean $\sqrt{1-\beta_t x_{t-1}}$ and variance $\beta_t \mathbf{I}$ is operated to produce x_t. In (1) and (2), x_0 gradually loses the distinguishable features as the noise level deepens. This recursion can be formulated explicitly using the reparameterization trick as follows:

$$q\left(x_t|x_0\right) = \mathcal{N}\left(x_t; \sqrt{\hat{\alpha}_t} x_0, (1-\hat{\alpha}_t)\mathbf{I}\right)$$
$$\Rightarrow x_t = \sqrt{\hat{\alpha}_t}\, x_0 + \sqrt{1-\hat{\alpha}_t} z_t \tag{3}$$

where $\hat{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$ and $z_t \sim \mathcal{N}(0, \mathbf{I})$. By (3), the latent noise x_t can be sampled at arbitrary time steps, which is further used to calculate the tractable objective loss $\mathcal{L}_t$.

2) Normality Memorizing: The process of normality memorizing involves specific mechanisms to effectively capture and retain the normal patterns during the forward noise encoding process. Fig. 3 shows the schematic illustration of the normality memorizing step. Denote the code memories as $\mathcal{M}$: $\{M_1, M_2, \ldots, M_T\}$, where the tth memory $M_t \in \mathbb{R}^{1\times K\times n}$ records prototypical patterns from the noise encoding in the tth step. We denote each query $M_t^k \in \mathbb{R}^{1\times 1\times n}$, where $k = 1, 2, \ldots, K$. The noise encoding result $X_t \in \mathbb{R}^{H\times W\times n}$ is output from a batch of images, where H, W, and n are height, width, and the size of training batches, respectively. The normality memorizing step serves two purposes: 1) transform the feature vector X_t^i of size $1\times 1\times n$ using the code memory M_t and 2) update the memory query M_t^k of size $1\times 1\times n$ using the feature map X_t.

As shown in Fig. 3, the cosine similarity between each encoded vector X_t^i and the code memory M_t is first calculated as follows:

$$w_t^{i,k} = \frac{\exp\left(\left(M_t^k\right)^T X_t^i\right)}{\sum_{k'=1}^{K}\exp\left(\left(M_t^{k'}\right)^T X_t^i\right)} \tag{4}$$

where $w_t^{i,k}$ is the corresponding weights after the softmax function. By integrating the memory queries with corresponding weights, the transformed feature vector $\hat{X}_t^i$ taking into account all normal patterns is obtained

$$\hat{X}_t^i = \sum_{k'=1}^{K} w_t^{i,k'} M_t^{k'}. \tag{5}$$

Applying (4) and (5) to all encoded vectors in X_t results in a transformed feature map $\hat{X}_t \in \mathbb{R}^{H\times W\times n}$, which is then added to X_t and fed into the next denoise sampling step. Similar to (4), we then compute the cosine similarity between each memory query M_t^k and the encoded feature map X_t as follows:

$$v_t^{i,k} = \frac{\exp\left(\left(M_t^k\right)^T X_t^i\right)}{\sum_{i'=1}^{H\times W} \exp\left(\left(M_t^k\right)^T X_t^{i'}\right)} \tag{6}$$

and normalize $v_t^{i,k}$ considering the feature indices for the corresponding features of M_t^k

$$v_t^{\prime i,k} = \frac{v_t^{i,k}}{\max_{i'\in D_t^k} v_t^{i',k}}. \tag{7}$$

With the normalized weights in (7), the query in code memory M_t is updated as follows:

$$M_t^k \leftarrow f\left(M_t^k + \sum_{i\in D_t^k} v_t^{\prime i,k} X_t^i\right) \tag{8}$$

where $f(\cdot)$ denotes the L2 norm, and D_t^k denotes the indices of the multiple feature vectors nearest to M_t^k according to (6).

In the normality memorizing step, a set of code memories obtained from the noise encoding step is integrated to establish consistent representations of normality for subsequent steps.

3) Denoise Memory-Suppress Sampling: The last denoise sampling is a reverse process approximating $q(x_{t-1}|x_t)$ as in DDPM. When $t = T$, image samples with high fidelity should be reconstructed from isotropic Gaussian noise. Unlike the first step, the reconstruction requires the knowledge of all previous gradients, which can only be obtained with a learning model. Thus, this step aims to train a subnetwork with learned parameters θ, thereby estimating $p_\theta(x_{t-1}|f_u(x_t))$ based on the normality memorizing function f_u. It is formulated with mean and variance as follows:

$$\begin{aligned} &p_\theta\ \left(x_{t-1}|f_u\left(x_t\right)\right) \\ &= \mathcal{N}\left(x_{t-1}; \mu_\theta\left(f_u\left(x_t\right), t\right), \Sigma_\theta\left(f_u\left(x_t\right), t\right)\right) \end{aligned} \tag{9}$$

where the mean function and the variance function are computed as in [19]

$$\begin{aligned} \mu_\theta\left(f_u\left(x_t\right), t\right) &= \frac{1}{\sqrt{\alpha_t}}\ \left(f_u(x_t') - \frac{1-\alpha_t}{\sqrt{1-\hat{\alpha}_t}} z_\theta\left(f_u\left(x_t\right), t\right)\right) \\ \Sigma_\theta\left(f_u\left(x_t\right), t\right) &= \exp\left(\boldsymbol{v}\log\beta_t + (1-\boldsymbol{v})\log\tilde{\beta}_t\right) \end{aligned} \tag{10}$$

where $\alpha_t = 1-\beta_t$, $\tilde{\beta}_t = \frac{1-\hat{\alpha}_{t-1}}{1-\hat{\alpha}_t}\ \beta_t$, and $\boldsymbol{v}$ is a mixing vector via model predicting. If we apply (9) in all T time steps, the trajectory from x_T to x_0 is:

$$p_\theta\ \left(x_{T:0}\right) = p(x_T\ ) \prod_{t\ =\ 1}^{T} p_\theta\left(x_{t-1}|f_u\left(x_t\right)\right). \tag{11}$$

Let the transformed features $\hat{x}_t = f_u\ (x_t)$. The objective of this module is to train the sampling function $p_\theta(x_{t-1}|\hat{x}_t)$ with the loss function defined as the variational lower bound loss

$$\begin{aligned} \mathcal{L}_{vlb} &= \sum_{t=0}^{T} \mathcal{L}_t \\ &= \mathbb{E}_{q(x_{0:T})}\left[\log\frac{q\left(x_{1:T}|x_0\right)}{p_\theta\left(\hat{x}_{T:0}\right)}\right] \\ &= \mathbb{E}_q\left[-\log p_\theta\left(x_0|\hat{x}_1\right)\ \rightarrow \mathcal{L}_0\right. \\ &\quad + \sum_{t>1} D_{KL}\left(q\left(x_{t-1}|x_t, x_0\right) \parallel p_\theta\left(x_{t-1}|\hat{x}_t\right)\right)\ \rightarrow \mathcal{L}_{t>1} \\ &\quad \left. + D_{KL}\left(q\left(x_T|x_0\right) \parallel p_\theta\left(x_T\right)\right)\right] \rightarrow \mathcal{L}_T \end{aligned} \tag{12}$$

where D_{KL} denotes the Kullback–Leibler divergence between two Gaussian distributions, which can be expressed in a closed form. The first term $\mathcal{L}_0$ is modeled using a separate decoder, which is derived from $\mathcal{N}(x_0; \mu_\theta(\hat{x}_1, 1), \Sigma_\theta(\hat{x}_1, 1))$ as in [18]. The second term $\mathcal{L}_{t>1}$ ensures that at each time step t, the estimation of $p_\theta(x_{t-1}|\hat{x}_t)$ is close to the true posterior of $q(x_{t-1}|x_t)$ as much as possible when conditioned on the input image. The last term $\mathcal{L}_T$ is constant since the encoding function q has no learnable parameters and x_T is a Gaussian noise. We can find in (12) that $\mathcal{L}_T$ does not depend on θ, implying that it can be ignored during training. Based on (12), an alternative loss function is employed to offer better reconstruction quality over $\mathcal{L}_{vlb}$

$$\mathcal{L}_s = \mathbb{E}_{t\sim[1:T],\ x_0\sim p(x_0), z_t\sim\mathcal{N}(0,\boldsymbol{I})}\ \left[\left\|z_t - z_\theta\left(\hat{x}_t, t\right)\right\|^2\right] \tag{13}$$

where $\mathbb{E}$ is the expected value and $z_\theta(\hat{x}_t, t)$ is the predicted noise in the sampling step t.

B. Contrastive Dissimilarity Network

The contrastive dissimilarity network uses the input image and the reconstructed image to calculate the correlation map and further predict the anomaly map. Inspired by [30], three components are involved in the contrastive dissimilarity network, including an encoder, a projector, and a fusion block as shown in Fig. 2.

1) Architecture: We utilize a pretrained visual geometry group (VGG) as the encoder to process both the original input x_0 and the reconstruction $p_\theta(x_0)$, resulting in two embedding vectors. We then project the two vectors into a larger space with a higher dimension $d = 1024$. It is simply a three-layer perceptron with batch normalization and ReLU activation and learns the invariance of the two images. The design of the vector projection is common and crucial in CRL-based methods, since it is capable of squeezing all invariant information and preventing model collapse. The output of the projector is denoted as follows:

$$\begin{aligned} h^A &= f_{E,P}\ \left(x_0\right) \in \mathbb{R}^d \\ h^B &= f_{E,P}(p_\theta\ \left(x_0\right)) \in \mathbb{R}^d. \end{aligned} \tag{14}$$

The final fusion block aims to compute the correlation map $C(h^A, h^B)$. Given the projector output, the correlation map is expected to be as close to the identity matrix as possible. For a batch size n, $C = \frac{\left(h^A\right)^T h^B}{n}\ \in \mathbb{R}^{d\times d}$ is the empirical correlation map. The diagonal elements of the target C (i.e.,

C_{ii}) are equal to 1, forcing the embeddings invariant to the reconstruction. The off-diagonal elements of the target C (i.e., C_{ij} for $i \neq j$) are equal to 0, which decorrelates the different components of embedding vectors and reduces the redundancy between h^A and h^B. With the fusion block, the network is directed to pay attention to high-dissimilar areas in the correlation map.

2) Objective Function: The objective function of the contrastive dissimilarity network encompasses multiple components aimed at achieving effective differentiation between the input image and its corresponding reconstruction. Additionally, it involves a pointwise correlation with the high-level embedding vectors. The objective can be expressed through three distinct terms, i.e., reconstruction loss $\mathcal{L}_{\text{rec}}$, projection feature loss $\mathcal{L}_{\text{pro}}$, and contrastive loss $\mathcal{L}_{\text{con}}$ as follows:

$$\begin{aligned}\mathcal{L}_d &= \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{pro}} + \mathcal{L}_{\text{con}} \\ &= \sum_i^n \|x_0 - p_\theta(x_0)\|_2 \\ &\quad + \lambda \sum_i^n \left\|h^{\mathrm{A}} - h^B\right\|_2 \\ &\quad + \sum_i (1 - C_{ii})^2 + \lambda_{\text{con}} \sum_i \sum_{i \neq j} C_{ij}^2 \end{aligned} \tag{15}$$

where $\mathcal{L}_{\text{rec}}$ measures the L2 distance between the original images and their reconstructions, ensuring a faithful representation. $\mathcal{L}_{\text{pro}}$ calculates the L2 distance between the embedding vectors h^{A} and h^B. $\mathcal{L}_{\text{con}}$ plays a crucial role in balancing the tradeoff between the invariance term and the redundancy reduction term. It consists of two components: 1) the sum of squared differences $1 - C_{ii}$ within the diagonal elements of the contrastive matrix C, which emphasizes the preservation of invariance and 2) the sum of squared differences C_{ij}^2 for all nondiagonal elements ($i \neq j$) of C, promoting redundancy reduction. $\mathcal{L}_{\text{con}}$ is the tradeoff parameter controlling the invariance term and the redundancy reduction term.

C. Training and Testing

1) Semisupervised Training: During the training process, the memory-suppress diffusion module and the contractive dissimilarity network are jointly optimized to minimize the sum of $\mathcal{L}_s$ and $\mathcal{L}_d$. The code memories are continuously updated to record the prototypical patterns of normal images. To speed up the sampling process, we adopt a multistep solution approximating the high-order derivatives as recommended in [21].

The training of contrastive dissimilarity networks also adopts a semisupervised manner that does not rely on negative pairs. It incorporates a quantification mechanism to assess the degree of abnormality, enabling the differentiation between normal and anomalous images.

2) Image-Level and Pixelwise Detection: A key aspect of ARLA lies in its ability to achieve high-fidelity reconstruction. We define a weighted dissimilarity score $\mathcal{E}(x)$ to express the RFOD at the image level

$$\varepsilon\ (x) = \sum_{i,j} W_{ij}\left(x_0, p_\theta(x_0)\right) \left\| x_0^{ij} - \left(p_\theta(x_0)\right)^{ij} \right\|_2 \tag{16}$$

where W_{ij} is the weight function defined as follows:

$$W_{ij}\left(x_0, p_\theta(x_0)\right) = \frac{1 - \exp\left(-\left\|x_0^{ij} - \left(p_\theta(x_0)\right)^{ij}\right\|_2\right)}{\sum_{i,j} 1 - \exp\left(-\left\|x_0^{ij} - \left(p_\theta(x_0)\right)^{ij}\right\|_2\right)}. \tag{17}$$

It determines the relative significance of each term and allows the ARLA to focus more on the regions with high dissimilarity. Meanwhile, a threshold ζ is empirically decided to minimize the errors corresponding to false alarms and miss-detected anomalies. A small $\mathcal{E}(x)$ indicates a normal input while a large $\mathcal{E}(x)$ exceeding ζ for an anomalous input.

Once the testing sample is determined as anomalous, a stacked pixelwise anomaly map is generated by merging a score distance map M_s and a feature distance map M_d along the depth dimension. The two maps are defined as follows:

$$\begin{aligned} M_s &= W_{ij}\ \left(x_0, p_\theta(x_0)\right)\left\|x_0^{ij} - \left(p_\theta(x_0)\right)^{ij}\right\|_2 \in \mathbb{R}^{H\times W} \\ M_d &= \omega \left\|F\left(x_0\right) - F\left(p_\theta(x_0)\right)\right\|_2^2 \end{aligned} \tag{18}$$

where M_s is derived from the dissimilarity score. M_d is derived from the encoded feature maps in a contrastive dissimilarity network. F denotes the last activation layer before feature flattening. ω is the mean squared error (MSE) weight to explore the effect of M_d.

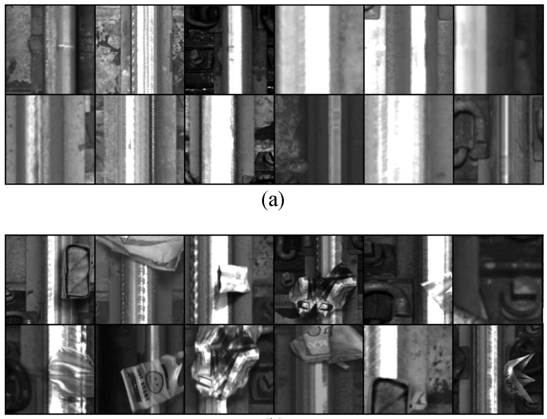
3) Architecture Training Complexity: This section provides an analysis of the complexity of the ARLA in training. As the ARLA is composed of a memory-suppress diffusion module and a contrastive dissimilarity network, its training complexity is described as follows. Let $g_1(\cdot)$ denote the complexity of memory-suppress operation, the complexity of the memory-suppress diffusion with K code memories within T steps over N training iterations is described as $f_{\text{MDM}} \in O(N \cdot T \cdot g_1(K)^T)$. Let $g_2(\cdot)$ denote the complexity of convolution operation and C_l denote the output channels of the lth layer, the complexity of contrastive learning process of dissimilarities within D layers over N training iterations is described as $f_{\text{CDN}} \in O(N \cdot \sum_l^D g_2(C_{l-1}C_l))$. Combining these two parts involves considering the composition and interactions of each part, such as the determination of anomalous samples at the image level.

IV. Experiments

In this section, we first introduce the railway dataset and setups employed during training and testing. Then we show some properties of ARLA and compare it with state-of-the-art methods on pixelwise RFOD. Finally, we present discussions on the components of ARLA.

A. Implementation Details

1) Dataset: The railway dataset used in our experiments was collected by the Hong Kong Metro Corporation via industrial cameras. A selection of collected samples is presented in Fig. 4.



(a)

(b)

Fig. 4. Collected samples from (a) D_{train} and (b) D_{test}.

The dataset comprises two main subsets: D_{train}, consisting of normal images intended for training, and D_{test}, which includes images with anomalies intended for testing.

All images are labeled at the image level. D_{train} consists of 6656 normal images. D_{test} comprises 541 anomalous images and 490 normal images. To select the optimal hyperparameters during training, a small portion of D_{test} is reserved as the validation set D_{val}, which consists of 120 anomalous images and 100 normal images. Apart from the image-level labels, pixelwise annotations as ground truths for training are not provided. The anomalies manifest themselves in the form of over ten different classes of foreign objects, such as phones, bags, and umbrellas. Before being fed into the ARLA model for training and testing, each image is resized to a resolution of 256×256 and normalized to values between 0 and 1. Furthermore, the images are converted to grayscale.

Note: The dataset will be available for sharing upon request via email after the expiration of the nondisclosure agreement, strictly for scientific research purposes.

2) Training Settings: The training process for the memory-suppress diffusion module involves utilizing D_{train} to learn the normal patterns and update the code memories. In this regard, a setting $T = 50$ is applied to the noise encoding and sampling steps based on experimental trials. Details will be disclosed in Section IV-C. The variance scheduler β_t is defined in the range of $[0.0001, 0.02]$ as recommended in [19]. In the contrastive dissimilarity network, the encoder and projector are initialized using Xavier initialization. The hidden dimension of the projector output is set to 1024. Grid searches are conducted to determine the optimal weights for the hyperparameters λ and ω, aiming to achieve the best performance based on D_{val}. Additionally, the value of λ_{con} is set to 5×10^{-3} according to [30].

The entire pipeline of ARLA is trained end-to-end using the hybrid loss objective composed of $\mathcal{L}_s$ and $\mathcal{L}_d$, with an AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.99$). We adopt the initial learning rate of 5×10^{-4} and the weight decay of 10^{-4}. The default training schedule is 300 k iterations, with the learning rate divided by 10 at 200 k and 250 k iterations. All modules are trained with a batch size of 24 on three NVIDIA GeForce RTX 2080 GPUs.

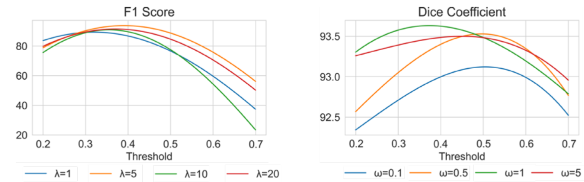


Fig. 5. Sensitivity analysis of weight parameters λ and ω.

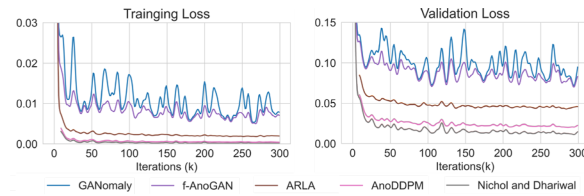


Fig. 6. Losses of benchmarks during training.

3) Inference Details: At the inference stage, the image is noise encoded and denoised back within 50 steps. Following the reconstruction, a dissimilarity score $\mathcal{E}$ and an anomaly map M are calculated. We determine the threshold ζ based on D_{val} such that any image with $\mathcal{E}$ larger than ζ is identified as anomalous. The batch size at the inference stage is set to 1.

4) Evaluation Metrics: Two distinct sets of evaluation metrics are employed to assess the detection performance of ARLA and benchmarking methods in our study. The first set encompasses commonly used metrics Precision, recall, and F1 score for image-level RFOD. The second set of evaluation metrics including dice coefficient and mean intersection over union (mIoU) evaluates the pixelwise detection performance. Moreover, the model complexity in terms of parameters, floating point operations per second (FLOPs), inference time, and FPS are also considered.

B. Results

1) Sensitivity Analysis: In this section, we perform experiments to evaluate the sensitivity of two hyperparameters: λ and ω, which govern the impact of projection feature loss and feature distance map on image-level and pixelwise detections, respectively. Fig. 5 describes the performance of the ARLA model in terms of the F1 score and dice coefficient based on different hyperparameter configurations. Results show that the ARLA model achieves its peak F1 score with $\lambda = 5$ and $\omega = 1$. Intriguingly, a remarkable consistency is observed in the optimal thresholds obtained across both groups of experiments, with minimal variance. A threshold $\zeta = 0.4$ is selected as a better performance is achieved based on the validation dataset D_{val}. These selected hyperparameters are subsequently employed for evaluation on D_{test}.

2) Comparison With Benchmarks: We compare our proposed ARLA against two groups of benchmarking methods. The first group includes two GAN-based methods: f-AnoGAN [13] and GANomaly [14]. The second group involves two DDPM-based methods: Nichol and Dhariwal [19] and AnoDDPM [20]. Fig. 6 shows the training loss and validation loss throughout the 300 k iterations. Default settings are adopted

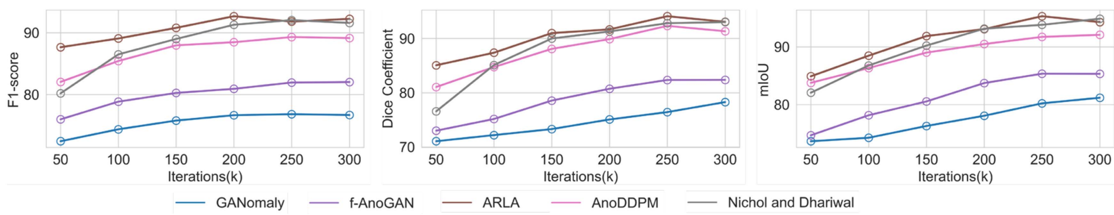


Fig. 7. Validation performances of ARLA and benchmarking methods.

TABLE II
COMPARISON WITH TWO GROUPS OF BENCHMARKING METHODS

Model	Image-level			Pixel-wise		Complexity			
	Precision	Recall	F1-score	Dice Coefficient	mIoU	#Parameters (M)	#FLOPs (G)	Inference Time (ms)	FPS
GANomaly	80.15	75.83	77.93	79.67	83.07	12.4	3.50	33.9	29
f-AnoGAN	83.80	81.61	82.19	82.49	87.95	16.7	4.27	46.0	22
Nichol and Dhariwal	93.28	91.37	92.32	93.14	96.74	33.7	5.56	82	12
AnoDDPM	91.49	90.26	90.87	92.08	93.38	45.5	6.06	103	9
ARLA	96.62	90.55	93.49	94.11	96.40	38.1	5.01	91	11

for all benchmarking methods. In Fig. 6, we first observe that the training losses of GANomaly and f-AnoGAN exhibit significant fluctuations after 30 k iterations while still showing a decreasing trend, whereas the training loss of the other three methods remains relatively stable. This indicates the necessity of continuing the training after the sudden drop in loss values. Second, the validation loss values of these five methods also show a gradual decline as the number of iterations increases, especially for GANomaly and f-AnoGAN. After 100 k iterations, their validation loss values experience noticeable fluctuations and then level off. Fig. 7 further illustrates the validation performance of ARLA and benchmarking methods. Results demonstrate a clear trend that the training iterations higher than 30 k lead to discernible improvements in performance metrics while the performance becomes stable after 300 k training iterations. Another finding is that most of the methods have reached optimal performance before the end of training. Therefore, during the inference stage, we exploit the model weights achieving the optimal validation performance to detect anomalies in D_{test}.

Quantitative comparisons based on D_{test} with respect to image-level and pixelwise metrics are represented in Table II. Initially, we compare ARLA against the two GAN-based methods. Within this subgroup, ARLA outperforms the previous methods, particularly in pixelwise detection, where the dice coefficient and mIoU exhibit significant improvements. Subsequently, we compare ARLA against the DDPM-based methods, which require pixelwise annotations during training. Given the imbalance between normal and anomalous samples, D_{train} and D_{val} are combined to form the training set for the DDPM-based methods. By simply evaluating the difference between reconstructions and the ground-truths, the anomalous images are identified with highlighted pixels via DDPM-based methods. Table II shows that our method achieves a noteworthy 2.6% improvement in F1 score while maintaining comparable performance in terms of dice coefficient and mIoU. Notably, our method involves fewer steps in the memory-suppress diffusion module than AnoDDPM, resulting in faster reconstruction progress during the inference stage. Finally, we summarize their FLOPs, total number of parameters, inference time, and FPS in Table II to provide a more intuitive representation of their complexity. The results indicate that GAN-based methods require significantly less time and computational resources compared to DDPM-based methods. Among the DDPM-based methods, our proposed ARLA demonstrates reasonable parameter volume and testing time while showing a certain advantage in terms of FLOPs.

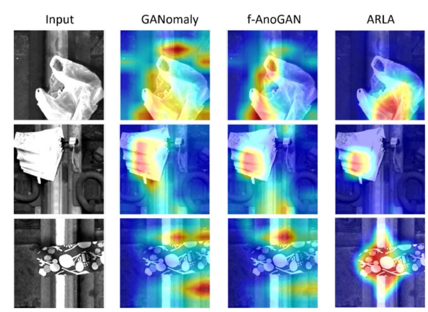


Fig. 8. Visualization of activation maps in neural networks.

3) Inference Visualization: We visualize the network weights of the contrastive dissimilarity network in ARLA as well as those of two GAN-based methods. Fig. 8 provides valuable insights into the ability of each method to capture and localize anomalous regions based on testing images. While ARLA predominantly focuses on highly concentrated anomalous regions, the GAN-based methods tend to highlight partially normal regions as well. This divergence in focus is reflected in the corresponding metrics, such as the dice coefficient and mIoU, as shown in Table II. The visualization comparison underscores the efficacy of ARLA in capturing dissimilarities between the input and its

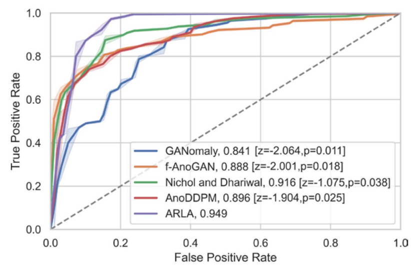


Fig. 9. ROC curves and p-values via DeLong's test.

TABLE III
ABLATION STUDIES ON EACH COMPONENT

(a)

T	F1	Dice Coefficient	mIoU
10	53.41	54.80	60.72
30	72.95	75.14	78.39
50	93.49	94.11	96.40
100	93.57	95.01	96.33

(b)

DDPM	Memory-suppression	F1	Dice Coefficient	mIoU
		81.07	84.50	84.73
✓		2.6	0.37	0.26
✓	✓	93.49	94.11	96.40

(c)

score dist. map	feature dist. map	F1	Dice Coefficient	mIoU
		79.88	81.26	80.43
✓		92.57	92.18	93.05
	✓	91.22	91.36	92.00
✓	✓	93.49	94.11	96.40

reconstruction, thereby facilitating precise pixelwise localization of anomalies.

4) Statistical Significance Test: Statistical significance tests are essential for confirming the effectiveness of the model. In this section, we incorporate DeLong's test [31] for comparing the receiver operating characteristic (ROC) curves and the area under ROC curves (AUROC) of all benchmarking methods. As shown in Fig. 9, the p-values of the four comparison groups are all smaller than a significance level of 0.05, indicating that the detection performance of benchmarking methods differs significantly from the proposed ARLA. When comparing ARLA with Nichol and Dhariwal, it shows a slightly higher p-value than the other groups. Adjusting the significance level will affect which groups are deemed significantly different. Nevertheless, a significance level of 0.05 is commonly used for conservative significance tests.

C. Discussions

1) Ablation Studies: We conduct comprehensive ablation studies on D_{test} to thoroughly analyze the individual contributions of the components in the proposed ARLA. The results are summarized in Table III, where the default settings of ARLA are marked in gray. First, we investigate the impact of sampling steps in the memory-suppress diffusion module. Table III(a) demonstrates that a sampling step value of 50 achieves a comparable F1 score, outperforming other groups by up to 40%. It is evident that ARLA requires a sufficient number of sampling steps to achieve the high-fidelity reconstruction of input images. However, excessive sampling steps result in minimal improvement in fidelity while consuming more time and affecting mIoU during the inference stage. As a result, we set the sampling step value as $T = 50$ to strike a balance between speed and quality in RFOD.

Furthermore, we validate the effectiveness of the memory-suppression strategy, as shown in Table III(b). When evaluating ARLA without the utilization of DDPM, normal images are directly fed into the contrastive dissimilarity network to learn the normal patterns between different images. Our findings indicate that the performance of ARLA significantly degrades in the absence of both DDPM and memory-suppression. Moreover, ARLA struggles to identify anomalous images when only DDPM is employed without the memory-suppression strategy, highlighting the necessity of incorporating memory-suppression in ARLA for effective detection performance.

In addition, we investigate different distance maps in Table III(c), including a score distance map M_s and a feature distance map M_d. The experimental results demonstrate that combining these two maps yields the best performance for pixelwise RFOD.

2) Data Augmentation: We also explore the promoting effect of data augmentation techniques on the ARLA framework. One set of experiments was based on traditional data augmentation, including geometric transformations (flipping, cropping, rotation, and stretching) and color space transformations (brightness change and saturation change), resulting in an enhanced version ARLA-AUG. Another set of experiments utilizes advanced data augmentation, Moment Exchange [32], leading to another model version named ARLA-ME. Additionally, we construct a larger dataset D_{aug} using traditional data augmentation techniques to examine the detection capabilities of these models when faced with more complex samples. To offer a visual representation of our findings, we present Fig. 10, which showcases samples from D_{aug} and elucidates their corresponding visualized detection results.

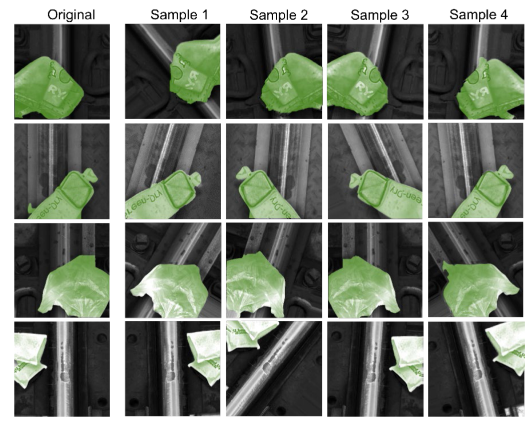


Fig. 10. Data augmentation samples and their visualized detection results from D_{aug}.

The assessment based on D_{test} and D_{aug} with quantitative insights are summarized in Table IV. We first find that even on a larger dataset D_{aug}, the detection prowess of the ARLA series

TABLE IV
EXPERIMENTAL RESULTS BASED ON D_{test} AND D_{aug}

	Model	Data augmentation		F1-score	Dice Coefficient	mIoU
		Traditional	Advanced			
D_{test}	ARLA			93.49	94.11	96.40
	ARLA-AUG	✓		93.28	94.50	96.42
	ARLA-ME	✓	✓	94.36	95.47	98.23
D_{aug}	ARLA			92.65	93.89	96.21
	ARLA-AUG	✓		93.00	94.19	96.58
	ARLA-ME	✓	✓	93.82	95.94	98.30

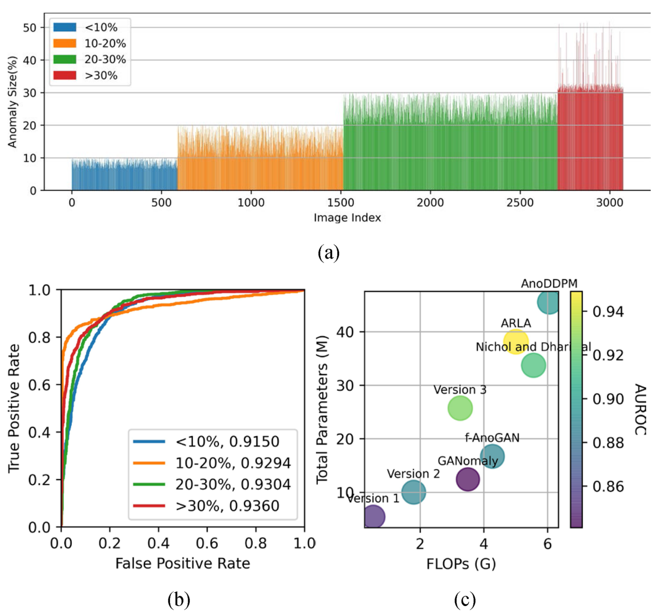


Fig. 11 (a) Histogram of all anomalous images in D_{aug}, where anomaly severity is categorized into four levels: <10%, 10%–20%, 20%–30%, and >30%. (b) ROC curves and AUROC of four anomaly severity levels. (c) Tradeoff between computational efficiency and detection performance based on D_{test}.

models remains impressively robust. Furthermore, our comparative analysis between ARLA-AUG and ARLA-ME yields noteworthy insights. While ARLA-AUG fails to manifest substantial improvements in both image-level and pixel-level detection, ARLA-ME emerges as a compelling alternative. Notably, in the context of D_{aug}, ARLA-ME achieves a remarkable 2.1% enhancement in both dice coefficient and mIoU. These results discernments distinctly underscore the scalability advantages inherent to the ARLA framework when faced with larger datasets and the intricacies of more complex railway environments.

3) Severity Differentiation: Extensive experiments are conducted to validate the robustness of ARLA in detecting anomalies in different severity levels. In these experiments, the anomaly severity is defined as the proportion of anomalous pixels in the entire image. Fig. 11(a) shows the distribution of anomaly sizes across 3138 anomalous images. The anomaly severity is categorized into four levels, <10%, 10%–20%, 20%–30%, and >30%. The majority of samples exhibit anomaly sizes ranging from 20-30%, indicating the prevalence of this size range in the dataset. Anomalies below 10% and between 10% and 20% occur less frequently, suggesting a lower incidence of smaller anomaly sizes. Anomalies exceeding 30% are relatively uncommon. Fig. 11(b) shows the discrimination ability of the ARLA across different anomaly severity levels. Impressively, ARLA demonstrates commendable detection performance even for small anomaly sizes, showcasing its versatility and robustness in detecting anomalies of varying magnitudes.

4) Computational Efficiency Improvement: We explore the data-free knowledge distillation method [33], [34], which derives lightweight model versions using the knowledge acquired from the original ARLA, to enhance the computational efficiency. Fig. 11(c) shows the tradeoff between computational efficiency (represented by FLOPs and total parameters) and detection performance (measured by AUROC). The lightweight versions, Versions 1–3, demanding fewer computational resources tend to exhibit lower AUROC. However, it is noteworthy that ARLA consistently maintains superior detection performance across all lightweight versions, underscoring its efficacy in anomaly detection tasks. Meanwhile, we notice that Version 3 emerges as a promising solution that strikes a balance between accuracy and efficiency for the RFOD problem. The enhanced performance of Version 3 underscores the potential of computational efficiency optimization strategies in advancing anomaly detection systems.

V. CONCLUSION

This study presented a comprehensive investigation of image-level and pixelwise RFOD without requiring anomaly samples in the context of anomalous railway scenes. We developed a novel approach, the ARLA, for classical RFOD tasks, which, in reality, was challenging to collect a sufficient amount of diverse anomaly samples. The ARLA included two complementary modules, a memory-suppress diffusion module, and a contrastive dissimilarity network. ARLA offered several advantageous features, such as less reliance on extensive anomalous data, ensuring a high-fidelity image reconstruction, and robustly performing RFOD with precise dissimilarity measurement. Through extensive experiments on a railway dataset, ARLA demonstrated superior performance compared to existing GAN-based and DDPM-based methods, positioning it as a potential solution for deployment in autonomous railway maintenance machines.

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